EE3110: Probability

Mon. 8am
Tue. 1pm
Thu. 11am
Fri 10am

courses.iitm.ac.in

ESB 127

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2 weeks: Tutorial

Book: Probability and Statistics
DeGroot & Schervish

Quiz 1, 25%
Quiz 2, 25%
Final 50%

Haricharan
haricharanb@gmail....

What is probability?

Grimmett & Stirzaker

Behind doors

1	2	3	Choice	Door opened		Switch wins
G_1	G_2	C	1	2	$\frac{1}{6} \times \frac{1}{3}$	✓
				3	$\frac{1}{6} \times \frac{1}{3}$	✓
			2	1	$\frac{1}{6} \times \frac{1}{3} \times \frac{1}{2}$	✗
			3	1	$\frac{1}{6} \times \frac{1}{3} \times \frac{1}{2}$	✗
			3	2	$\frac{1}{6} \times \frac{1}{3} \times \frac{1}{2}$	✗

$\frac{2}{3}$

Always show a goat. If 2, pick 1 uniformly at random.

G_1	G_2	C	1	2	$\frac{1}{6} \times \frac{1}{3} \times \frac{1}{2}$	✓
				3	$\frac{1}{6} \times \frac{1}{3} \times \frac{1}{2}$	✗
			2	1	$\frac{1}{6} \times \frac{1}{3} \times \frac{1}{2}$	✓
				3	$\frac{1}{6} \times \frac{1}{3} \times \frac{1}{2}$	✗
			3	1	"	✗
				2	"	✗

$\frac{1}{3}$

Pick 1 of 2 uniformly at random always.